DECISION 521 – Decision Analytics & Modeling

Fuqua School of Business

Spring Term 1, 2026

Galileo Optical Instrument Ltd

Part A: Due Class 8 (ungraded)

Part B: Due Class 9 (graded)

Assignment: Galileo Optical Instrument needs to analyze its purchasing decision for eyepieces of binoculars after Typhoon Lekima disrupted its original supplier. We will analyze this case in two steps.

Part A: In the first part, you should use the spreadsheet model (available on Canvas, under class 8) and ranges provided to develop a tornado chart that identifies the important uncertainties in the problem. Be sure you can explain any surprising findings in this analysis. Submit an electronic copy (one per team) on Canvas by deadline. The main deliverable is this electronic copy of the tornado chart(s) along with *a brief discussion (one or two paragraphs) summarizing your findings in this analysis*. **You need only submit one write-up per group**; this will not be formally graded. We will discuss these in class 8.

Part B:

In the second part, you will be required to model the uncertainties identified as important in part A. In part B of the case, you will be given additional information about the uncertainties. You should then develop a simulation model to answer the key question of the case: **How many eyepieces from the Kunshan supplier should Galileo order?** Your answer should discuss if there are any tradeoffs between risk and (expected) profit for different order quantities.

Each team should prepare a 10-15-minute PowerPoint presentation describing their analysis and recommendations. Prepare the presentation as if you were presenting to Pinpin, who has basic knowledge of simulation and is already *familiar with Part A of the analysis*. In each section, I will randomly select one or more groups to give their presentations to the class.

Deliverables for part B:

- Submit an electronic copy of your **presentation and simulation model(s)** using the Assignment tool in Canvas. I only need one electronic submission per team; please note *all of the team members* in the text box when you submit in Canvas.
- If there are comments you would make in your presentation but did not put in your slides, please include these comments in the presentation using the "Speaker notes" feature of PowerPoint.
- Please consider adding comments in your Python notebook next to your code or your analysis if you deem appropriate. You should also keep copies of these materials as you may find them useful for the class discussion.

Your assignment grade will be based on the quality of your analysis (e.g., correctness, completeness and appropriateness) as well as the quality of your presentation. Please take this as a somewhat open-ended exercise: if you have any interesting findings for Pinpin, please include them in your presentation.

IMPORTANT NOTE: Hints for Part B are provided on the back of this page.

Here are some important hints that would help you as you work on Part B of the case.

1. In Part A of the case, we use tornado analysis to identify some important uncertainties. For Part B, you should focus **solely on modeling these important uncertainties** based on the new information provided in Part B.

2. When it comes to fitting distributions to 10-50-90 percentiles, you may want to take advantage of the `fit_10_50_90` function from `model_uncertainty.py`. Note that in addition to 10-50-90s, the `fit_10_50_90` function also requires a minimum and maximum. For the sake of consistency, please consider setting the minimum to be 0.5*(10th percentile) and maximum to be 1.5*(90th percentile), unless there is a natural candidate for minimum and maximum.

3. When fitting distributions using the 10th, 50th, and 90th percentiles, you may find the `fit_10_50_90` function in `model_uncertainty.py` (posted under Class 8) helpful. In addition to the 10–50–90 percentiles, this function also requires a minimum and a maximum value. For consistency, I recommend setting the minimum to 0.5*(10th percentile) and the maximum to 1.5*(90th percentile).

4. There is NO need to round various quantities into integers (i.e., number of eyepieces) in your model, because rounding should not have any significant effect to your analysis.

5. Similar to the overbooking and retirement planning exercise, for this case, you will need to model a decision variable in your simulation model.

6. You may assume that Galileo's order volume from Kunshan is restricted to multiples of 1,000 eyepieces.

Galileo Optical Instrument Ltd. (A)[1]

Nantong, Jiangsu Province China

On the morning of August 10th, 2019, Lin Pinpin, the CEO of Galileo Optical Instrument Ltd, sat down in her office on Binjiang Park Road, having just read a memo from her purchasing team. After a cup of Longjing tea, Pinpin stood up to stare at the Yangtze River through her office window. The scenery of the great river provided Pinpin with a sense of peace that allowed her to focus on the decisions at hand.

The memo was about a supplier for Galileo Optical Instrument Ltd., which owns a manufacturing plant in Taizhou, Zhejiang province. Two days ago, the supplier's factory was just hit by a major flood as a result of Typhoon Lekima (see Exhibit E for Lekima's event map), causing severe damages. It was unclear when its production could resume. The supplier was responsible for producing eyepieces for a specific type of binocular, that Galileo is scheduled to produce for one of its major U.S. clients. Because of the supplier's production disruption, Galileo started to look at other alternatives for the production of eyepieces to fulfill its commitment.

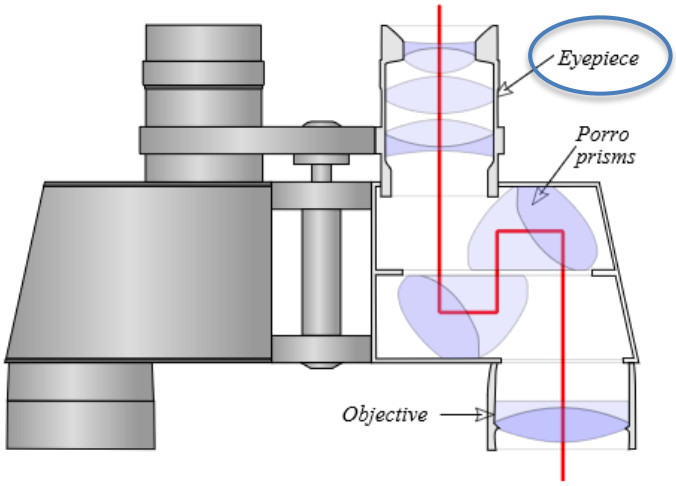

Figure 1: Illustration of a typical binocular design. (created by Antilived on Wikimedia Commons)

The authors of the memo are Wang Ziteng (Z.T. Wang), the firm's purchasing director, and Candace Shi, one of the firm's purchasing managers. Both Z.T. and Candace were hired by Pinpin after she became the CEO in 2018. Candace was a personal friend of Pinpin since they were both

[1] This case was developed by Yehua Wei and Peng Sun, Fuqua School of Business, Duke University in August 2020, and revised in February 2022 and January 2023. It was based on a consulting visit and various discussions with managers and the CEO of a manufacturing firm in China. To maintain the anonymity of the firm, all names used in the case are fictitious. In addition, the numbers presented the case are developed for classroom purposes and do not reflect actual business information of the firm.

toddlers. Their family friendship dates back to the last generation. Candace had recently graduated from the Fuqua School of Business at Duke University, and was recruited by Pinpin to bring more quantitative modeling and analysis to Lin's purchasing group. Z.T. was an experienced purchasing manager, who worked at a Japanese owned manufacturing firm near Nantong for eleven years before joining Galileo. Based on the forecast of Typhoon Lekima, Z.T. had predicted the possibility of a supplier disruption several days before it actually took place. In the meantime, Z.T. had used his network to identify a new supplier in Kunshan, in Jiangsu Province, that can meet the specification of the eyepieces Galileo needs.

Despite the new potential supplier in Kunshan, Pinpin did not wish to cancel the order from the supplier in Taizhou. The Taizhou supplier had been supplying Galileo for over fifteen years and always produced high quality lenses in a cost effective and timely manner. The supplier has been hit hard by the flood and will need all business it can get to stay afloat after it resumes production. Pinpin believed that it's important for Galileo to maintain a tight relationship with its key suppliers and decided to continue to proceed with the original order. More specifically, Galileo Optical Instrument will purchase any amount of eyepieces the Taizhou supplier is able to produce and deliver after resuming production, up to the original order quantity, by the deadline, October 11th, the latest date that Galileo need to start to assemble the binoculars in order to meet its own delivery deadline.

As Pinpin read through the memo, she noticed that the Kunshan supplier charges a significantly higher price than the Taizhou supplier. If the Taizhou supplier can somehow produce all the eyepieces Galileo needs by the deadline, Galileo would not need to order anything from the Kunshan supplier. However, there is considerable risk associated with such a strategy. If Galileo does not receive enough eyepieces by October 11th, it would not meet all of its client's demands. In addition, as orders for other parts of the binoculars have already been made, Galileo will be stuck with all those parts in its warehouse because of missing eyepieces.

Pinpin decided to call Z.T. and Candace to inform them about her decision on the Taizhou supplier. Pinpin's intuition was that they would also need eyepieces from the Kunshan supplier, but she was willing to let Z.T. and Candace work out the exact number. The memo noted that the Kunshan supplier requested the order quantity by noon of Sunday, August 11th, hence Z.T. and Candace will have to figure out the order quantity quickly. Pinpin noted that the order number is straightforward once they found out how much the Taizhou supplier can produce, but the uncertainties about the Taizhou supplier made the decision tricky. “This could be an excellent learning experience for the team to operate under uncertainty,” Pinpin thought.

Company Background

Galileo Optical Instrument is located in Nantong, a city located near the exit of the Yangtze River, approximately 80 miles north of Shanghai. Nantong is considered as one of the birthplaces to China’s modern manufacturing. The industrialization of Nantong took off in the early 20th century, when Zhang Jian, a prominent entrepreneur and educator, led the effort to build a series of modern factories, schools, as well as China’s first museum. Nantong’s status as a

manufacturing hub was further solidified in 1984, when it was selected as one of the 14 coastal cities opened to overseas investments, as the People's Republic of China started opening up the country's economy to the rest of the world.

In 1989, Galileo Optical Instrument Ltd was founded by Lin Ruhai as a family-owned business. At the time, it was one of the first privately owned companies manufacturing optical instruments in the People's Republic of China. Started as a manufacturer of inexpensive optical toys, the Galileo Optical grew steadily over time and had become one of the leaders in manufacturing optical instruments. By 2018, it had over 500 employees, owned over 100 patents, and opened sales offices in Asia, Europe, and North America.

Galileo's Change of Management

In 2018, Lin Ruhai decided to transition to a part-time role at Galileo, while transferring the CEO role to his daughter, Lin Pinpin. Before becoming Galileo's CEO, Pinpin studied as an MBA in the U.S. and worked in sales management at one of the largest multinational technology companies. At the time of transition, Galileo's growth was reaching a plateau: it was facing more business competition; had trouble keeping its profit margin as wages continued to rise; and started a downward trend on its manufacturing efficiency. After Pinpin became the CEO, she realized that in order for Galileo to remain competitive, it would transition to the more profitable mid to high-end optical market. Pinpin knew that this transition required not only advanced machineries but also more systematic and scientific management throughout the firm. As a result, Pinpin aggressively hired a new generation of young and talented managers. It was around that time when Z.T. and Candace joined Galileo. Historically, most of Galileo's suppliers were either owned or recommended by friends and relatives of her family. After Z.T. joined Galileo, his team worked with Pinpin to start a more quantitative approach in selecting their suppliers by identifying ways to measure suppliers' reliability, quality, and technical capability. Throughout this process, Galileo was able to reduce its costs by 4%, but more importantly, selected suppliers that are capable of delivering reliable parts for Galileo's new generation of optical instruments.

The Binocular Order

In July 2019, Pinpin's sales team was able to secure an order from one of its major U.S. clients for up to 16,000 binoculars. This order was considered as an important step for Galileo in transitioning into the binocular market with higher profit margins. The order required Galileo to deliver the binoculars by November 15th, 2019, and Galileo would be paid at 22 U.S. dollars for each binocular it delivers. Using an exchange rate of ¥7 per dollar, $22 is equivalent to ¥154.[2] In addition, Galileo's client mentioned that there might be future orders if the binocular is well received on the market. After the agreement, Galileo quickly ordered from its suppliers all of the parts needed for assembling 16,000 binoculars, including 32,000 of the eyepieces (each binocular requires two eyepieces) from the Taizhou supplier.

[2] All monetary values in this case, other than the $22 sales price, are in Renminbi or RMB, the Chinese currency with the symbol ¥.

Suppliers from Taizhou and Kunshan

The supplier from Taizhou agreed to produce 32,000 eyepieces for Galileo at ¥5 per eyepiece. The supplier initially planned to start its production of the eyepieces on August 19th, but due to the flood, it was clear that the supplier from Taizhou could not start its production on time. The municipality for the location of the supplier's manufacturing plant estimated that it would take two more days to resume electricity and pump the water out of the region. After that, an engineer will be sent to the manufacturing plant to assess the damage to its manufacturing machines. The supplier believed that in the most likely scenario, where some or all machines at the plant would be substantially damaged but fixable, it would take approximately five weeks from August 12th (the upcoming Monday) for the plant to resume the production. Once the production starts, the supplier will be able to produce up to 1000 eyepieces per day. However, it is also possible that the damage scenario is less severe, where all of the tools at the plant are still functional; or the damage scenario is more severe, where some or all machines are damaged beyond repair and need to be replaced.

Z.T. was able to get a quote on the eyepiece from the potential supplier from Kunshan. The Kunshan supplier has a capacity of producing up to 15,000 eyepieces by October 11th, and charges Galileo ¥8 per piece.. After some negotiations, Z.T. also agreed to the supplier's term on accepting an order that is higher than 15,000. In this case, the supplier argued that it would have pay overtime to their employees and delaying production on some existing orders. Therefore, they need to charge an additional twenty-five percent premium on each eyepiece above 15,000 (equates to ¥10 for each eyepiece ordered above 15,000).

Production Costs for the Binoculars

For each binocular, Galileo pays ¥90 for all other parts aside from the eyepieces. Galileo has already ordered all other parts under the plan of producing 16,000 binoculars. These other parts are scheduled to be delivered to Galileo before the scheduled assembly starting date, October 11th. Due to technicalities and limitations of their operational process, Galileo does not accept part shipments after the assembly starting date. If there are not enough eyepieces to assemble all of the 16,000 binoculars, then the other parts will stay in Galileo's warehouse and either used in other products or discarded. Candace estimated that overall, the parts that are unused for the binoculars could be salvaged at an approximate rate of 70% of its original ordering price. In addition, Candace estimated that the labor cost for assembling each binocular is about ¥22.5, and the salvage value for each unused eyepiece is ¥3.

Ordering from the Kunshan Supplier

As Pinpin made her decision with the Taizhou supplier, she asked Z.T. and Candace to determine the number of eyepieces to order from the Kunshan supplier. Pinpin encouraged Z.T. and Candace to make the decision based on a scientific approach using quantitative analysis. Candace volunteered to build a quantitative model to assist Z.T. with the order decision, as she realized that the problem at hand would make use of materials she learned from her MBA study.

To gain some initial understanding of the problem, Candace started with an Excel calculator that computes Galileo's profit given an order amount to the Kunshan supplier (see Exhibits A, B1, and B2). However, Candace quickly realized that the calculator was not a reliable predictor for Galileo's profit, as some of the numbers used in the calculator are based on very rough estimates. To develop a range for the parameters used in the calculator, Candace worked with Z.T. and other personals in Galileo to estimate each of the uncertain parameters not only for its base-case but also for the low and high scenarios (see Exhibit C). Candace believed that these numbers provide a reasonable estimate on the 10-50-90 range of the parameters.

Kunshan Order quantity (# of eyepieces)	0
Eyepeices per binocular	2
Revenue per unit (USD)	22
Total demand (# of binoculars)	16000
Time to delivery (days)	95
Kunshan supplier price per eyepiece (RMB)	8
Kunshan premium price (RMB)	10
Kunshan regular capacity (# of eyepieces)	15000
Taizhou supplier resume in (days)	35
Taizhou supplier daily capacity (# of eyepieces)	1000
RMB per USD	7
Cost for other parts (RMB)	90
Labor cost per binocular (RMB)	22.5
Salvage value per eyepiece (RMB)	3
Salvage Value for Other Parts (RMB)	63
Time to assemble and test after receiving eyepieces (days)	35
Taizhou supplier price per eyepiece (RMB)	5
Total Profit	¥299,250.00

Exhibit A. Basic inputs to the model

	A	B	C	D
1			**Cash Flow Analysis**	
2				
3	Eyepiece Orders			
4		Eyepieces from Taizhou		25000
5		Eyepieces from Kunshan		0
6				
7	Binoculars Sold			
8		Total demand		16000
9		Total eyepieces		25000
10		Total Sold		12500
11				
12	Revenue & Future Credits (in RMB)			
13		Revenue from binoculars		¥1,925,000.00
14		Total salvage value from eyepieces		¥0.00
15		Total salvage value from other parts		¥220,500.00
16	Total credits			¥2,145,500.00
17				
18	Operations costs (in RMB)			
19		Paid to Taizhou		¥125,000.00
20		Paid to Kunshan (regular price)		¥0.00
21		Paid to Kunshan (premium price)		¥0.00
22		Costs of other parts		¥1,440,000.00
23		Labor costs		¥281,250.00
24	Total costs			¥1,846,250.00
25				
26	Total profit			¥299,250.00

Exhibit B1. Profit Calculation

	A	B	C	D
1			**Cash Flow Analysis**	
2				
3	Eyepiece Orders			
4		Eyepieces from Taizhou		=MIN(Total_demand*2,MAX(0,Time_to_delivery-T_starting_day-Time_assemble_test)*T_daily_cap
5		Eyepieces from Kunshan		=Order_quant
6				
7	Binoculars Sold			
8		Total demand		=Total_demand
9		Total eyepieces		=(D4+D5)
10		Total Sold		= MIN(D9/2, D8)
11				
12	Revenue & Future Credits (in RMB)			
13		Revenue from binoculars		=D10*Revenue_per_unit*Conversion_rate
14		Total salvage value from eyepieces		=(D9-2*D10)*Salvage_value_eyep
15		Total salvage value from other part:		=(Total_demand-D10)*Salvage_value_other
16	Total credits			=SUM(D13:D15)
17				
18	Operations costs (in RMB)			
19		Paid to Taizhou		=-D4*T_price
20		Paid to Kunshan (regular price)		=-K_price*MIN(K_capacity, D5)
21		Paid to Kunshan (premium price)		=-MAX(D5-K_capacity,0)*K_premium_price
22		Costs of other parts		=-Cost_other_parts*Total_demand
23		Labor costs		=-Labor_cost*D10
24	Total costs			=SUM(D19:D23)
25				
26	Total profit			=D16+D24

Exhibit B2. Profit Calculation (with formula)

			Low	Base	High
Taizhou supplier resume in (days)	35	(5 weeks from August 12)	28	35	42
Taizhou supplier daily capacity (# of eyepieces)	1000		500	1000	1000
RMB per USD	7		6.7	7	7.3
Cost for other parts (RMB)	90		89.5	90	90.5
Labor cost per binocular (RMB)	22.5		18	22.5	25.5
Salvage value per eyepiece (RMB)	3		2	3	5
Salvage Value for Other Parts (RMB)	63		60	63	70
Time to assemble and test after receiving eyepieces (days)	35	(October 11 to November 15)	33	35	35
Taizhou supplier price per eyepiece (RMB)	5		4.5	5	5

Exhibit C. The Low-Base-High estimates of the input parameters

Exhibit D. Important upcoming dates for Galileo

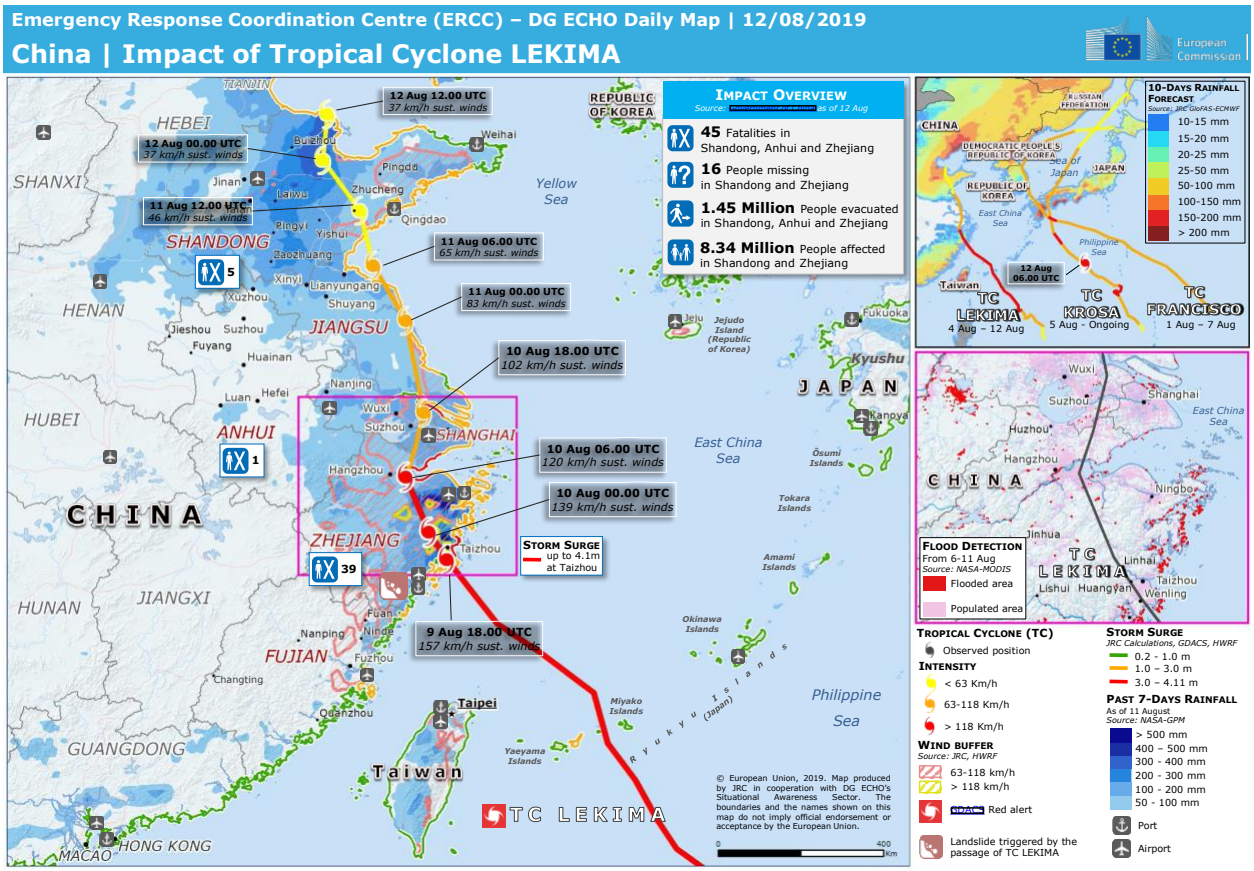

Exhibit E. Event Map for Typhoon Lekima
Source: www.gdacs.org